Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 3B nr. 2.6

$$f(x) = \frac{1}{x^2 + 2x}$$
$$f'(x) = \frac{-(2x-2)}{(x^2+2x)^2}$$
$$f''(x) = \frac{(2 \cdot 1 + 0)(x^2+2x)^2 - (2x-2) \cdot 2(x^2+2x)(2x+2 \cdot 1)}{(x^2+2x)^4}$$
$$f''(x) = \frac{2(2x-2)(2x+2)}{(x^2+2x)^3} - \frac{2}{(x^2+2x)^2}$$
$$f''(x) = \frac{2(3x^2+6x+4)}{x^3(x+2)^3}$$
$$f'''(x) = \frac{2(x^3(x+2)^3(6x+6) - (3x^2(x+2)^3 + x^3 \cdot 3(x+2)^2(1+0))(3x^2+6x+4))}{x^6(x+2)^6}$$
$$f'''(x) = \frac{6(3x^2+6x+4)}{x^2(x+2)^3} - \frac{6(3x^2+6x+4)}{x^3(x+2)^4} + \frac{2(6x+6)}{x^3(x+2)^3}$$
$$f'''(x) = -\frac{24(x^3+3x^2+4x+2)}{x^4(x+2)^4}$$

References